WebSocket — Complete Notes (Beginner to Advanced)

> Complete WebSocket learning notes for Python Backend Developers.
>
> Topics: Fundamentals, HTTP vs WebSocket, Handshake, Architecture, FastAPI, Django Channels, Redis, Authentication, Security, Scaling, Projects and Interview Questions.

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1. What is WebSocket?

WebSocket is a communication protocol that provides persistent, full-duplex communication between a client and a server over a single TCP connection.

In simple words:

WebSocket allows the client and server to communicate with each other in real time without creating a new HTTP request for every message.

Key Features

- Full-duplex communication.
- Persistent connection.
- Real-time data exchange.
- Low communication overhead after connection establishment.
- Supports text and binary messages.
- Works over TCP.
- Supports communication in both directions.

Example

Suppose you are using a real-time chat application.

1. User A sends a message.
2. Server receives the message.
3. Server forwards the message to User B.
4. User B receives the message immediately.

This type of communication can be implemented using WebSocket.

Real-Life Examples

- WhatsApp-like chat applications.
- Facebook Messenger.
- Live notifications.
- Online multiplayer games.
- Stock market dashboards.
- Live delivery tracking.
- Collaborative editing applications.

---

2. Why Do We Need WebSocket?

Traditional HTTP follows a request-response communication model.

For example:

1. Client sends a request.
2. Server processes the request.
3. Server sends a response.
4. The request-response cycle completes.

If the client wants updated information, it generally needs to send another request.

This is not ideal for real-time applications.

Problems with Traditional HTTP

- Repeated requests.
- Additional communication overhead.
- Delayed updates without polling.
- Inefficient for frequent real-time communication.

WebSocket Solution

WebSocket establishes a persistent connection between the client and server.

After establishing the connection:

- Client can send messages anytime.
- Server can send messages anytime.
- Both can communicate independently.
- No repeated HTTP handshake is required for each message.

---

3. HTTP vs WebSocket

Feature	HTTP	WebSocket
Communication	Request-Response	Full-Duplex
Connection	Request-based	Persistent
Direction	Client initiates requests	Both directions
Real-time	Requires additional techniques	Native support
Overhead	Repeated HTTP requests	Lower after handshake
Protocol	HTTP/HTTPS	WS/WSS
Transport	TCP (HTTP/1.1 and HTTP/2)	TCP
Best for	REST APIs	Real-time applications

HTTP Example

Client request:

```http
GET /api/users/ HTTP/1.1
Host: example.com
```

Server response:

```json
{
    "id": 1,
    "name": "Faruk"
}
```

WebSocket Example

Client sends:

```text
Hello Server
```

Server responds:

```text
Hello Client
```

The same connection can remain open for additional messages.

---

4. WebSocket Protocol

WebSocket is standardized by RFC 6455.

There are two common WebSocket URL schemes.

4.1 WS

`ws://` represents an unencrypted WebSocket connection.

Example:

```text
ws://localhost:8000/ws/chat/
```

4.2 WSS

`wss://` represents a WebSocket connection protected by TLS encryption.

Example:

```text
wss://example.com/ws/chat/
```

Difference Between WS and WSS

WS	WSS
Unencrypted	Encrypted
No TLS protection	Uses TLS
Suitable for local testing	Recommended for production
Data can be intercepted more easily	Protects data in transit

Always use WSS in production.

---

5. How WebSocket Works

WebSocket communication happens in several stages.

Step 1: Client Initiates Connection

The browser creates a WebSocket connection.

```javascript
const socket = new WebSocket(
    "ws://localhost:8000/ws"
);
```

Step 2: HTTP Handshake

The client sends an HTTP Upgrade request to the server.

Step 3: Server Accepts Connection

The server responds with:

```http
HTTP/1.1 101 Switching Protocols
Upgrade: websocket
Connection: Upgrade
```

Step 4: Connection Established

Now the connection is upgraded to WebSocket.

Step 5: Data Exchange

Client and server exchange messages over the established connection.

Step 6: Connection Closed

Either the client or server can close the connection.

---

6. WebSocket Handshake

The WebSocket handshake is the initial process used to establish a WebSocket connection.

It starts with an HTTP request.

Client Handshake Request

```http
GET /ws/chat/ HTTP/1.1
Host: localhost:8000
Upgrade: websocket
Connection: Upgrade
Sec-WebSocket-Key: example-key==
Sec-WebSocket-Version: 13
```

Important Headers

Header	Description
Upgrade	Requests protocol upgrade
Connection	Indicates connection upgrade
Sec-WebSocket-Key	Random handshake key
Sec-WebSocket-Version	WebSocket protocol version
Sec-WebSocket-Accept	Server-generated handshake response
Origin	Identifies the request origin

Server Handshake Response

```http
HTTP/1.1 101 Switching Protocols
Upgrade: websocket
Connection: Upgrade
Sec-WebSocket-Accept: generated-value
```

What is HTTP 101?

HTTP status code `101 Switching Protocols` means the server has accepted the protocol upgrade.

After the handshake, communication continues using WebSocket frames.

---

7. WebSocket Architecture

A typical WebSocket application contains the following components:

1. Client.
2. WebSocket Server.
3. Application Logic.
4. Database.
5. Message Broker.
6. Load Balancer.

Example Architecture

```text
                CLIENTS
                   |
          ---------------------
          |         |         |
        User A    User B    User C
          |         |         |
          ---------------------
                    |
               WebSocket
                    |
             LOAD BALANCER
                    |
```

```
          ---------------------
          |                   |
       Server A            Server B
          |                   |
          ---------------------
                    |
                 REDIS
                    |
                DATABASE
```

Component Responsibilities

Component	Responsibility
Client	Sends and receives messages
WebSocket Server	Manages connections
Application Logic	Processes business logic
Database	Stores persistent information
Redis	Enables distributed communication
Load Balancer	Distributes incoming connections

---

8. Important WebSocket Terminologies

Term	Meaning
Handshake	Initial connection establishment
Full-Duplex	Both sides can send data
Persistent Connection	Connection remains open
Frame	Basic WebSocket transmission unit
Message	Application data sent between peers
Ping	Connection health check
Pong	Response to ping
Close Frame	Connection termination signal
Heartbeat	Periodic connection health checking
Origin	Source origin of the connection
Subprotocol	Application-level protocol negotiation
Backpressure	Handling data faster than the receiver can process
Broadcast	Sending a message to multiple clients
Consumer	Django Channels connection handler
Channel Layer	Communication system between consumers

---

9. WebSocket Events

JavaScript provides several WebSocket events.

9.1 open

Triggered when the connection is successfully established.

```javascript
socket.onopen = () => {
    console.log("Connected!");
};
```

9.2 message

Triggered when a message is received.

```javascript
socket.onmessage = (event) => {
    console.log("Received:", event.data);
};
```

9.3 error

Triggered when an error occurs.

```javascript
socket.onerror = (error) => {
    console.log("Error:", error);
};
```

9.4 close

Triggered when the connection closes.

```javascript
socket.onclose = () => {
    console.log("Disconnected!");
};
```

---

10. WebSocket Connection Lifecycle

WebSocket has four ready states.

State	Value	Meaning
CONNECTING	0	Connection is being established
OPEN	1	Connection is open
CLOSING	2	Connection is closing
CLOSED	3	Connection is closed

Example:

```javascript
console.log(socket.readyState);
```

```
if (socket.readyState === WebSocket.OPEN) {
    socket.send("Hello Server");
}
```

Connection Lifecycle

```text
CONNECTING
     |
     v
    OPEN
     |
     v
   CLOSING
     |
     v
   CLOSED
```

---

11. WebSocket with JavaScript

JavaScript provides a built-in WebSocket API.

Create Connection

```javascript
const socket = new WebSocket(
    "ws://localhost:8000/ws"
);
```

Complete Example

```javascript
const socket = new WebSocket(
    "ws://localhost:8000/ws"
);

socket.onopen = () => {
    console.log("WebSocket connected");
};

socket.onmessage = (event) => {
    console.log("Server:", event.data);
};

socket.onerror = (error) => {
    console.error("WebSocket error:", error);
};

socket.onclose = () => {
```

````
    console.log("WebSocket disconnected");
};

function sendMessage(message) {
    if (socket.readyState === WebSocket.OPEN) {
        socket.send(message);
    }
}
```

### Send Text Message

```javascript
socket.send("Hello Server");
```

### Send JSON Message

```javascript
socket.send(
    JSON.stringify({
        type: "chat",
        message: "Hello Faruk"
    })
);
```

### Receive Message

```javascript
socket.onmessage = (event) => {
    const data = JSON.parse(event.data);

    console.log(data.message);
};
```

---

# 12. WebSocket with Python

Python supports WebSocket through different libraries.

Popular options:

- FastAPI WebSocket.
- Django Channels.
- websockets library.
- Starlette WebSocket.

### Install WebSockets Library

```bash
pip install websockets
````

```

### Simple WebSocket Server

```python
import asyncio
from websockets.asyncio.server import serve


async def handler(websocket):
    async for message in websocket:
        print("Received:", message)

        await websocket.send(
            f"Server received: {message}"
        )


async def main():
    async with serve(
        handler,
        "localhost",
        8765
    ):
        await asyncio.Future()


asyncio.run(main())
```

This example uses the modern `websockets` asyncio server API.

---

13. WebSocket with FastAPI

FastAPI provides built-in WebSocket support.

Install Dependencies

```bash
pip install fastapi uvicorn
```

Project Structure

```text
websocket_project/
│
├── main.py
└── requirements.txt
```

main.py
```

```python
from fastapi import FastAPI, WebSocket

app = FastAPI()


@app.websocket("/ws")
async def websocket_endpoint(
    websocket: WebSocket
):
    await websocket.accept()

    try:
        while True:
            data = await websocket.receive_text()

            await websocket.send_text(
                f"Server received: {data}"
            )

    except Exception:
        print("Connection disconnected")
```

Run Server

```bash
uvicorn main:app --reload
```

WebSocket URL:

```text
ws://127.0.0.1:8000/ws
```

Code Explanation

1. Import WebSocket

```python
from fastapi import WebSocket
```

Imports the WebSocket class.

2. Create WebSocket Route

```python
@app.websocket("/ws")
```

Defines a WebSocket endpoint.

3. Accept Connection

```python
await websocket.accept()
```

Accepts an incoming WebSocket connection.

4. Receive Message

```python
data = await websocket.receive_text()
```

Receives a text message from the client.

5. Send Message

```python
await websocket.send_text(data)
```

Sends a text message to the client.

6. Disconnect Handling

```python
except Exception:
    print("Connection disconnected")
```

For production, prefer catching `WebSocketDisconnect` specifically.

Better FastAPI Example

```python
from fastapi import (
    FastAPI,
    WebSocket,
    WebSocketDisconnect
)

app = FastAPI()



@app.websocket("/ws")
async def websocket_endpoint(
    websocket: WebSocket
):
    await websocket.accept()

    try:
        while True:
```

```
            message = await websocket.receive_text()

            await websocket.send_text(
                f"Echo: {message}"
            )

    except WebSocketDisconnect:
        print("Client disconnected")
```

JSON Communication

```python
from fastapi import FastAPI, WebSocket

app = FastAPI()


@app.websocket("/ws")
async def websocket_endpoint(
    websocket: WebSocket
):
    await websocket.accept()

    while True:
        data = await websocket.receive_json()

        message = data.get("message")

        await websocket.send_json({
            "type": "response",
            "message": message
        })
```

---

14. WebSocket with Django Channels

Django Channels extends Django to support protocols such as WebSocket using ASGI.

What is Django Channels?

Django Channels is a framework that enables asynchronous communication in Django applications.

It supports:

- WebSocket.
- Long-running connections.
- Asynchronous consumers.
- Channel layers.
- Group broadcasting.

Install

```bash
pip install channels channels-redis
```

Project Structure

```text
project/
│
├── config/
│   ├── settings.py
│   ├── asgi.py
│   └── urls.py
│
├── chat/
│   ├── consumers.py
│   ├── routing.py
│   └── models.py
│
└── manage.py
```

Step 1: settings.py

```python
INSTALLED_APPS = [
    "channels",
    "django.contrib.admin",
    "django.contrib.auth",
    "django.contrib.contenttypes",
    "django.contrib.sessions",
    "django.contrib.messages",
    "django.contrib.staticfiles",
]

ASGI_APPLICATION = "config.asgi.application"
```

Step 2: consumers.py

A Consumer is responsible for handling WebSocket connection events and messages.

```python
import json

from channels.generic.websocket import (
    AsyncWebsocketConsumer
)


class ChatConsumer(AsyncWebsocketConsumer):
```

```
    async def connect(self):
        await self.accept()

        await self.send(text_data=json.dumps({
            "message": "Connected successfully"
        }))

    async def disconnect(self, close_code):
        print("Disconnected")

    async def receive(self, text_data):
        data = json.loads(text_data)

        message = data.get("message")

        await self.send(text_data=json.dumps({
            "message": message
        }))
```

Step 3: routing.py

```python
from django.urls import re_path

from .consumers import ChatConsumer

websocket_urlpatterns = [
    re_path(
        r"ws/chat/$",
        ChatConsumer.as_asgi()
    ),
]
```

Step 4: asgi.py

```python
import os

from django.core.asgi import get_asgi_application
from channels.routing import ProtocolTypeRouter, URLRouter

from chat.routing import websocket_urlpatterns


os.environ.setdefault(
    "DJANGO_SETTINGS_MODULE",
    "config.settings"
)

django_asgi_app = get_asgi_application()
```

```
application = ProtocolTypeRouter({
    "http": django_asgi_app,

    "websocket": URLRouter(
        websocket_urlpatterns
    ),
})
```

Important Consumer Methods

Method	Description
connect()	Handles connection
accept()	Accepts connection
disconnect()	Handles disconnection
receive()	Handles incoming message
send()	Sends message

---

15. WebSocket Groups and Broadcasting

A WebSocket Group allows multiple connections to receive the same event.

Example

Suppose 10 users are connected to a chat room.

When one user sends a message, all users in the room should receive that message.

This is called broadcasting.

Group Methods

Method	Description
group_add()	Adds connection to a group
group_discard()	Removes connection from a group
group_send()	Sends event to group

Django Channels Group Example

```python
import json

from channels.generic.websocket import (
    AsyncWebsocketConsumer
)


class ChatConsumer(AsyncWebsocketConsumer):

    async def connect(self):
```

```
        self.room_group_name = "chat_room"

        await self.channel_layer.group_add(
            self.room_group_name,
            self.channel_name
        )

        await self.accept()

    async def disconnect(self, close_code):

        await self.channel_layer.group_discard(
            self.room_group_name,
            self.channel_name
        )

    async def receive(self, text_data):

        data = json.loads(text_data)

        message = data.get("message")

        await self.channel_layer.group_send(
            self.room_group_name,
            {
                "type": "chat_message",
                "message": message
            }
        )

    async def chat_message(self, event):

        await self.send(
            text_data=json.dumps({
                "message": event["message"]
            })
        )
```

How Broadcasting Works

```text
                 CHAT ROOM
                     |
          ------------------------
          |          |           |
        User A     User B      User C
          |
      Sends Message
          |
      group_send()
          |
      Channel Layer
```

```
          |
     ------------------
     |        |        |
   User A   User B   User C
   Receives Receives Receives
   Message  Message  Message
```

---

16. WebSocket with Redis

Redis is an in-memory data store that can be used as a message broker and channel layer backend.

Why Redis?

Suppose an application has two WebSocket servers.

```text
User A                  User B
   |                        |
Server A                  Server B
   |                        |
   ----------- REDIS --------
```

User A is connected to Server A.

User B is connected to Server B.

When User A sends a message, Redis can distribute the event so that Server B can deliver it to User B.

Install Redis Channel Layer

```bash
pip install channels-redis
```

settings.py

```python
CHANNEL_LAYERS = {
    "default": {
        "BACKEND": "channels_redis.core.RedisChannelLayer",

        "CONFIG": {
            "hosts": [
                ("127.0.0.1", 6379)
            ],
        },
    },
}
```

```

Redis must be running.

### Redis Responsibilities

- Distributed message delivery.
- Cross-server communication.
- Channel layer backend.
- Real-time event distribution.

Important: Redis channel layers are not a replacement for persistent message storage. Store important chat messages in a database.

---

# 17. WebSocket Authentication

Authentication ensures that only authorized users can establish or use protected WebSocket connections.

### Authentication Methods

1. Session Authentication.
2. JWT Authentication.
3. Token Authentication.
4. Custom Authentication Middleware.

### Django Channels Session Authentication

```python
from channels.auth import AuthMiddlewareStack
from channels.routing import (
    ProtocolTypeRouter,
    URLRouter
)

from django.core.asgi import get_asgi_application

application = ProtocolTypeRouter({

    "http": get_asgi_application(),

    "websocket": AuthMiddlewareStack(
        URLRouter(websocket_urlpatterns)
    ),

})
```

Check User Authentication

```python
class ChatConsumer(AsyncWebsocketConsumer):
```
```

```
    async def connect(self):

        user = self.scope["user"]

        if user.is_anonymous:
            await self.close()
            return

        await self.accept()
```

JWT Authentication

JWT can be validated during the WebSocket handshake using custom middleware or secure token validation.

Example conceptual flow:

```text
Client
   |
   | JWT Token
   |
WebSocket Handshake
   |
JWT Validation
   |
   |---- Invalid Token ---> Reject
   |
Valid Token
   |
Authenticated Connection
```

Important: Do not assume that a JWT is automatically validated by Django Channels. You must implement and configure the authentication mechanism.

---

18. WebSocket Security

Security is extremely important for production WebSocket applications.

18.1 Use WSS

Always use encrypted WebSocket connections in production.

```text
wss://example.com/ws/chat/
```

18.2 Origin Validation

Validate the Origin header to prevent unauthorized cross-origin WebSocket connections.

18.3 Authentication

Verify the user's identity before allowing access to protected connections.

18.4 Authorization

Check whether the authenticated user has permission to access a room or resource.

18.5 Input Validation

Validate all incoming messages.

18.6 Message Size Limits

Prevent excessively large messages.

18.7 Rate Limiting

Limit the number of messages a user can send within a period.

18.8 Avoid Sensitive Data

Never send passwords, secret keys, or unnecessary sensitive information through WebSocket messages.

Security Checklist

- [ ] Use WSS.
- [ ] Validate Origin.
- [ ] Authenticate users.
- [ ] Authorize room access.
- [ ] Validate incoming data.
- [ ] Set message size limits.
- [ ] Implement rate limiting.
- [ ] Handle disconnections.
- [ ] Protect against unauthorized subscriptions.
- [ ] Monitor suspicious activity.

---

19. Polling vs Long Polling vs SSE vs WebSocket

These are different techniques for delivering updates between clients and servers.

19.1 Polling

Client repeatedly sends requests to the server at fixed intervals.

Example:

```javascript
setInterval(() => {
```

```
    fetch("/api/notifications/");

}, 5000);
```

Advantages:

- Easy to implement.
- Works with normal HTTP APIs.

Disadvantages:

- Repeated requests.
- Unnecessary server load.
- Updates may be delayed.

19.2 Long Polling

Client sends an HTTP request.

The server holds the request until new data is available or a timeout occurs.

After receiving a response, the client sends another request.

Advantages:

- Better than frequent polling for event-driven updates.
- Works with standard HTTP.

Disadvantages:

- Repeated requests still occur.
- More complex connection management.

19.3 Server-Sent Events (SSE)

SSE provides a persistent HTTP connection through which the server sends events to the client.

It is one-way communication.

```text
Server -------> Client
```

Useful for:

- Live news feeds.
- Notifications.
- Server-side status updates.

19.4 WebSocket

WebSocket supports bidirectional communication.

```text
Client <------> Server
```

Useful for:

- Chat applications.
- Multiplayer games.
- Live collaboration.
- Real-time tracking.

Comparison Table

Feature	Polling	Long Polling	SSE	WebSocket
Protocol	HTTP	HTTP	HTTP	WS/WSS
Direction	Client requests	Client requests	Server to client	Bidirectional
Persistent	No	Temporary	Yes	Yes
Real-time	Limited	Better	Yes	Yes
Best for	Periodic updates	Event updates	Live feeds	Interactive apps

---

20. WebSocket Ping and Pong

Ping and Pong are WebSocket control frames.

Ping

One endpoint sends a Ping frame to check whether the connection is alive.

Pong

The other endpoint responds with a Pong frame.

Why Are They Important?

- Detect dead connections.
- Keep connections healthy.
- Identify network interruptions.
- Support connection monitoring.

Heartbeat Flow

```text
Client                 Server
  |                       |
  |------ PING ---------->|
  |                       |
  |<----- PONG -----------|
  |                       |
  |    Connection Alive   |
```

```

A heartbeat can also be implemented at the application level using normal messages.

Important: Browser JavaScript does not expose direct control over WebSocket Ping and Pong frames. These are generally managed by the WebSocket implementation or server.

---

# 21. Error Handling and Reconnection

Network failures can cause WebSocket disconnections.

A robust application should handle unexpected disconnections.

### Basic Reconnection Example

```javascript
let socket;

function connectWebSocket() {

    socket = new WebSocket(
        "ws://127.0.0.1:8000/ws"
    );

    socket.onopen = () => {
        console.log("Connected");
    };

    socket.onclose = () => {

        console.log("Disconnected");

        setTimeout(() => {
            connectWebSocket();
        }, 3000);

    };

    socket.onerror = (error) => {
        console.log("WebSocket error", error);
    };
}

connectWebSocket();
```

Production Reconnection Strategy

Use:

- Exponential backoff.
- Maximum retry attempts.
```

- Random jitter.
- Connection timeout.
- Authentication refresh.
- Reconnection status tracking.

Exponential Backoff Example

```javascript
let retryDelay = 1000;
const maxDelay = 30000;

function scheduleReconnect() {

    const delay = retryDelay;

    retryDelay = Math.min(
        retryDelay * 2,
        maxDelay
    );

    setTimeout(connectWebSocket, delay);
}
```

Reset the retry delay after a successful connection.

---

22. WebSocket Performance Optimization

For production applications:

1. Use asynchronous I/O.
2. Avoid blocking operations.
3. Use Redis for distributed communication.
4. Keep messages small.
5. Implement connection limits.
6. Use efficient serialization.
7. Handle slow consumers.
8. Implement backpressure.
9. Monitor active connections.
10. Configure proper timeouts.
11. Optimize database queries.
12. Avoid unnecessary broadcasts.

Why Avoid Blocking Operations?

WebSocket servers often handle many concurrent connections.

A blocking operation can prevent the event loop from processing other tasks efficiently.

Use asynchronous operations where supported.

---

23. WebSocket Scaling

Scaling means handling more users and more simultaneous connections.

Single Server Architecture

```text
Clients
   |
   |
WebSocket Server
   |
Database
```

Problem: A single server may become a bottleneck.

Multiple Server Architecture

```text
                CLIENTS
                   |
              LOAD BALANCER
                   |
          ---------------------
          |                   |
       Server A            Server B
          |                   |
          ---------------------
                    |
                 REDIS
                   |
                DATABASE
```

Important Scaling Components

Component	Purpose
Load Balancer	Distributes connections
ASGI Servers	Handle asynchronous connections
Redis	Cross-server communication
Database	Persistent data storage
Monitoring	Tracks system health

Important Considerations

- Long-lived connection management.
- Load balancer WebSocket support.
- Connection draining during deployments.
- Graceful shutdown.
- Redis availability.
- Database connection management.

- Horizontal scaling.

---

24. Real-World WebSocket Use Cases

1. Chat Application

- One-to-one messaging.
- Group chat.
- Online status.
- Typing indicators.
- Read receipts.

2. Notification System

- Instant notifications.
- Read/unread status.
- Notification history.
- User-specific notifications.

3. Live Dashboard

- Live analytics.
- Real-time metrics.
- Server monitoring.
- Stock price updates.

4. Live Tracking

- GPS location updates.
- Delivery tracking.
- Driver location.
- Real-time order status.

5. Multiplayer Games

- Real-time player actions.
- Game state synchronization.
- Live score updates.

6. Collaborative Applications

- Shared document editing.
- Live cursor positions.
- Real-time collaboration.

---

25. WebSocket Best Practices

Backend Best Practices

- Use asynchronous consumers where appropriate.

- Keep consumers lightweight.
- Validate incoming messages.
- Authenticate every connection.
- Check authorization for every protected action.
- Use Redis for distributed communication.
- Store important messages in a database.
- Handle disconnect events.
- Configure connection limits.
- Monitor connection counts.

Frontend Best Practices

- Check connection state before sending.
- Handle connection errors.
- Implement reconnection.
- Show connection status.
- Validate received data.
- Avoid duplicate event handlers.
- Clean up connections when appropriate.

Production Best Practices

- Use WSS.
- Configure load balancing.
- Use health checks.
- Implement heartbeat monitoring.
- Configure timeouts.
- Use structured logging.
- Monitor memory and CPU.
- Test concurrent connections.
- Handle graceful shutdown.

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26. WebSocket Interview Questions and Answers

Beginner Level

Q1. What is WebSocket?

Answer:

WebSocket is a communication protocol that provides persistent, full-duplex communication between a client and a server over a single TCP connection.

Q2. What is the difference between HTTP and WebSocket?

Answer:

HTTP follows a request-response model, whereas WebSocket provides persistent bidirectional communication.

Q3. What is a WebSocket handshake?

Answer:

It is the initial HTTP Upgrade process used to establish a WebSocket connection.

Q4. What is the difference between WS and WSS?

Answer:

WS is unencrypted, whereas WSS uses TLS encryption.

Q5. What is full-duplex communication?

Answer:

Full-duplex communication allows both client and server to send and receive data independently at the same time.

Q6. What is a persistent connection?

Answer:

A persistent connection remains open so that multiple messages can be exchanged without establishing a new connection each time.

Q7. What is HTTP 101?

Answer:

HTTP 101 Switching Protocols indicates that the server has accepted a protocol upgrade request.

Intermediate Level

Q8. What is a WebSocket Group?

Answer:

A WebSocket Group allows multiple connections to receive the same broadcast event.

Q9. Why do we use Redis with Django Channels?

Answer:

Redis provides a channel layer that enables communication between multiple application instances.

Q10. What is a Consumer in Django Channels?

Answer:

A Consumer handles WebSocket connection lifecycle events and incoming and outgoing messages.

Q11. What is the difference between Consumer and View?

Answer:

A Django View normally handles HTTP requests and returns HTTP responses. A Channels Consumer handles protocol events and can manage long-lived connections.

Q12. Can WebSocket use JWT authentication?

Answer:

Yes. JWT can be validated during the handshake using custom authentication middleware or another secure token validation mechanism.

Q13. What happens when a WebSocket disconnects?

Answer:

The connection closes, and the server should release associated resources and remove the connection from relevant groups.

Q14. What is Broadcasting?

Answer:

Broadcasting means sending a message or event to multiple connected clients.

Q15. What is a Channel Layer?

Answer:

A channel layer is a communication abstraction that allows different consumers or application instances to exchange events.

Advanced Level

Q16. How do you scale WebSocket applications?

Answer:

Use multiple ASGI server instances, a shared channel layer such as Redis, load balancing with WebSocket support, and proper connection management.

Q17. How do you secure WebSocket connections?

Answer:

Use WSS, origin validation, authentication, authorization, input validation, message limits, and rate limiting.

Q18. What is the difference between WebSocket and SSE?

Answer:

WebSocket supports bidirectional communication, while SSE provides server-to-client event streaming.

Q19. What is a heartbeat mechanism?

Answer:

A heartbeat periodically checks whether a connection is still alive.

Q20. What is backpressure?

Answer:

Backpressure is a mechanism for handling situations where data is produced faster than the receiver can process it.

Q21. Why is Redis useful in distributed WebSocket applications?

Answer:

Redis allows events to be distributed between different application instances, making cross-server communication possible.

Q22. What is the difference between WebSocket Message and Frame?

Answer:

A frame is a protocol-level unit of transmission. A message is the complete application-level data, which may consist of one or multiple frames.

Q23. Does WebSocket automatically provide authentication?

Answer:

No. Authentication must be implemented by the application.

Q24. Does WebSocket automatically store messages?

Answer:

No. WebSocket is a communication protocol. Persistent message storage must be implemented using a database or another storage system.

Q25. What is WebSocket Origin Validation?

Answer:

It verifies the origin of a connection request to help prevent unauthorized cross-origin connections.

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27. Practical WebSocket Projects

Project 1: Real-Time Chat Application

```
Level: Beginner to Intermediate.

### Features

- User registration.
- User login.
- One-to-one messaging.
- Group chat.
- Message timestamps.
- Message history.
- Online/offline status.
- Typing indicator.

### Technology Stack

- Python.
- FastAPI or Django Channels.
- PostgreSQL.
- Redis.
- JavaScript.

## Project 2: Real-Time Notification System

Level: Intermediate.

### Features

- Instant notifications.
- User-specific notifications.
- Read/unread status.
- Notification history.
- Notification count.

### Technology Stack

- Django.
- Django Channels.
- Redis.
- PostgreSQL.

## Project 3: Live Delivery Tracking

Level: Advanced.

### Features

- Driver location updates.
- Customer live tracking.
- Delivery status updates.
- Live GPS coordinates.
- Real-time map updates.

### Technology Stack
```

- FastAPI.
- WebSocket.
- Redis.
- PostgreSQL.
- JavaScript Maps API.

Project 4: Real-Time Dashboard

Level: Advanced.

Features

- Live analytics.
- Active user count.
- Live server status.
- Real-time data visualization.
- Dashboard updates.

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28. Complete WebSocket Learning Roadmap

Step	Topic	Priority
01	HTTP vs WebSocket	Essential
02	WebSocket Handshake	Essential
03	WebSocket Protocol	Essential
04	JavaScript WebSocket API	Essential
05	FastAPI WebSocket	Essential
06	Django Channels	Essential
07	Groups and Broadcasting	Essential
08	Redis Channel Layer	Important
09	Authentication	Important
10	Security	Important
11	Error Handling	Important
12	Reconnection	Important
13	Heartbeat	Important
14	Performance Optimization	Advanced
15	Scaling	Advanced
16	Production Deployment	Advanced
17	Real-World Projects	Essential

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29. Official Documentation

WebSocket Protocol

- RFC 6455: https://datatracker.ietf.org/doc/html/rfc6455

JavaScript

- MDN WebSocket API: https://developer.mozilla.org/en-US/docs/Web/API/WebSocket

FastAPI

- WebSocket Documentation: https://fastapi.tiangolo.com/advanced/websockets/

Django Channels

- Official Documentation: https://channels.readthedocs.io/en/latest/

Redis

- Official Documentation: https://redis.io/docs/latest/

ASGI

- ASGI Specification: https://asgi.readthedocs.io/en/latest/

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30. Final Summary

WebSocket is an important technology for building real-time applications.

Important Concepts to Remember

1. WebSocket provides full-duplex communication.
2. WebSocket uses a persistent TCP connection.
3. The connection starts with an HTTP Upgrade handshake.
4. HTTP 101 means Switching Protocols.
5. WS is unencrypted; WSS uses TLS.
6. FastAPI supports WebSocket natively.
7. Django Channels enables WebSocket in Django.
8. Consumers handle WebSocket events.
9. Groups support broadcasting.
10. Redis enables distributed communication.
11. Authentication and authorization are essential.
12. WebSocket does not automatically store messages.
13. Ping and Pong help monitor connections.
14. Reconnection improves reliability.
15. Load balancing and Redis help scale applications.

Recommended Learning Strategy

For Python Backend Development:

1. Understand HTTP and WebSocket fundamentals.
2. Practice WebSocket using JavaScript.
3. Build a simple WebSocket server using FastAPI.
4. Learn Django Channels.
5. Implement group broadcasting.
6. Integrate Redis.
7. Implement authentication and authorization.
8. Build a complete real-time chat application.
9. Deploy the application using ASGI and WSS.

```
**Goal:** Become comfortable building secure, scalable, production-ready real-time backend applications using Python, FastAPI, Django Channels, and Redis.

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**End of WebSocket Complete Notes.**
```